ASSIGNMENT – 4

Introduction to Symmetric Cryptography | BCA309T

Q1. Explain the basic principles of symmetric cryptography and Kerckhoffs's principle.

Symmetric cryptography uses the same secret key to encrypt and decrypt data. Plaintext is the original message; ciphertext is the encrypted message; the encryption algorithm converts plaintext to ciphertext, and the decryption algorithm reverses it using the key. Kerckhoffs's principle says a system should remain secure even if its design is public; the secret key must provide the security.

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Plaintext → [Encryption algorithm + secret key] → Ciphertext
Plaintext  [Decryption algorithm + same secret key]  Ciphertext
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Q2. Distinguish stream and block ciphers. Explain confusion, diffusion, and block modes.

A stream cipher encrypts data one bit or byte at a time, while a block cipher encrypts fixed-size blocks. Confusion hides the relationship between the key and ciphertext; diffusion spreads the influence of each plaintext bit across many ciphertext bits.

- ECB: Encrypts each block independently; simple, but reveals repeated patterns.
- CBC: Combines each block with the previous ciphertext block; hides patterns better, but needs an initialization vectorand sequential processing.
- CFB: Uses previous ciphertext to make a block cipher act like a stream cipher; useful for streaming, but errors can affectfollowing data.
- OFB: Generates a keystream; transmission errors do not spread much, but the IV/nonce must never be reused with thesame key.

CTR: Encrypts counter values to create a keystream; fast and parallelizable, but counter/nonce reuse is dangerous.

Q3. Trace the historical development of symmetric encryption.

Early systems used classical methods such as the Caesar cipher. During World War II, the Enigma machine provided mechanical encryption. Claude Shannon's information-theory work formalized ideas such as confusion and diffusion. Feistel networks influenced modern block cipher design. DES was adopted in 1977, but its short key later became vulnerable to brute-force search. Triple DES improved security by applying DES multiple times. Other algorithms included IDEA, Blowfish, and RC4. An open competition led to the selection of Rijndael as AES in 2000; AES became a standard in 2001.

Q4. Explain Caesar, monoalphabetic, Playfair, and Vigenère ciphers. Why is frequency analysis useful?

- Caesar cipher: Shifts each letter by a fixed number. It is easy to use but easy to break by trying the few possible shifts.
- Monoalphabetic substitution: Replaces each letter with a fixed substitute. It has many possible keys, but letter-frequencypatterns remain.
- Playfair cipher: Encrypts pairs of letters using a 5×5 letter table. It hides single-letter frequencies better, but can still beattacked using digraph patterns.
- Vigenère cipher: Uses a repeating keyword to apply different shifts. It resists simple frequency analysis better thanCaesar, but repeating-key patterns can reveal the key.

Frequency analysis works when letter patterns remain consistent; it is less effective when substitutions vary across positions.

Q5. Explain transposition, product ciphers, and the one-time pad.

A rail fence cipher writes letters in a zigzag pattern and reads them row by row. A columnar transposition writes text in a grid and reads columns in a key-defined order. Transposition changes letter positions but not the letters themselves. A product cipher combines substitution and transposition, giving both confusion and diffusion; modern block ciphers build on this idea. A one-time pad uses a truly random key as long as the message, used only once. It is theoretically unbreakable when used correctly, but sharing and storing such long random keys is impractical.

Q6. Describe the structure of DES and one round.

DES is a symmetric Feistel block cipher. It encrypts 64-bit blocks using an effective 56-bit key (8 additional bits are used for parity). After an initial permutation, the block is split into 32-bit left and right halves and processed through 16 rounds. Each round expands the right half from 32 to 48 bits, XORs it with a 48-bit round key, passes the result through S-boxes to 32

bits, and applies a P-box permutation. The halves are combined and a final permutation is applied. The key schedule creates sixteen 48-bit round keys from the main key.

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One DES round: L(i+1) = R(i);  R(i+1) = L(i) XOR F(R(i), K(i))
F: Expansion → XOR round key → S-boxes → P-box
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Q7. Discuss DES weaknesses, Double DES, and Triple DES.

DES has a useful avalanche effect: a small input change causes many output bits to change. However, its 56-bit key is too short, making brute-force attacks practical with modern hardware. Differential and linear cryptanalysis are also important analysis techniques. Double DES applies DES twice but is vulnerable to a meet-in-the-middle attack, so it does not provide the expected security increase. Triple DES applies DES three times; 2-key and 3-key variants improve security, but it is slower than AES and is now largely legacy.

Q8. Explain AES, its transformations, and compare it with DES.

AES is a symmetric substitution-permutation block cipher with a 128-bit block size. It supports 128-, 192-, and 256-bit keys, using 10, 12, and 14 rounds respectively. Each normal round uses SubBytes (substitutes bytes), ShiftRows (shifts rows), MixColumns (mixes bytes within columns), and AddRoundKey (XORs with a round key). The final round omits MixColumns. Key expansion generates round keys from the original key. Decryption applies inverse transformations in reverse order. Compared with DES, AES has a larger block size, much longer keys, stronger modern security, and generally better performance in current systems.